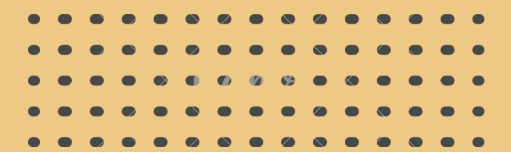


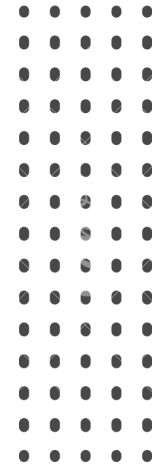


MEASURING WTP FOR HOUSING

How does accessibility to tube stations affect housing value?



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Introduction + Literature Review (I)

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Literature Review (II) + Methodology

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Analysis + Conclusion



LITERATURE REVIEW

Baum-Snow & Kahn (2000)

The effects of new public projects to expand urban rail transit

Context :

- five major US cities that upgraded their rail transit systems in the 1980s : Boston, Atlanta, Chicago, Portland and Washington DC

Approach :

- Cross-sectional **census tract-level** data
- Compares tracts before and after rail development
- Distance as proximity measure

Strengths +Limitations !

X Omitted Variable Bias

X Reverse Causality

X Correlated Effects

Findings :

- a **small** behavioral response of incumbent residents toward increased commuting by public transit, and a small capitalization of transit infrastructure into housing prices and rents.
- New transit's benefits are not uniformly distributed.



LITERATURE REVIEW

Billings (2011) :

Estimating the value of a new transit option

Context :

- Charlotte's LRT System

Approach :

- Difference-in-Differences using abandoned and completed LRT corridors
- single-family homes, condos, commercial units
- Uses both categorical bins (0.5 mile, 1 mile) and continuous distance
- Checks both neighborhood effects AND price gradients

Strengths +Limitations !

✓ Omitted Variable Bias

✓ Reverse Causality

X Correlated Effects

Findings:

- 4% price increase within 1 mile for single-family homes (but not within 0.5 mile)
- Stronger effects for condos and new commercial development
- Suggests LRT value may come more from redevelopment than accessibility



LITERATURE REVIEW

Gibbons & Machin (2005) :

Valuing rail access using transport innovations

Context :

- construction of new stations in London in the late 1990s

Approach :

- quasi-experimental, difference-in-difference approach to avoid the biases inherent in cross-sectional valuation work
- Uses repeat sales microdata
- Measures exact distance to nearest station (continuous)
- Time-differencing variables over time (unobserved characteristics)

Strenghts +Limitations !

✓ Omitted Variable Bias

✓ Reverse Causality

X Correlated Effects

Findings:

- households value rail access and that these valuations are large compared to the valuations of other local amenities (like local primary school performance)
- house prices rose over the period by 9.3 percentage points more in places affected by these transport infrastructure changes, relative to places that were unaffected



LITERATURE REVIEW

Bowes and Ihlanfeldt (2001):

Measuring the effects of rail stations on property values in Atlanta

Context :

- Focus on residential properties near rail stations in Atlanta

Approach :

- Hedonic price regression model
- Included housing and neighbourhood variables (income, education, crime rate)
- Examined both positive and negative externalities of rail proximity
- Analysed non-linear effects of distance (up to 3 miles from station)

Limitations !

- ✓ Omitted Variable Bias
- X Context Specificity
- X Endogeneity (Reverse Causality)

Findings:

- Property prices rise as distance to the nearest station decreases but fall again when houses are too close
- Strongest positive effects for homes between 0.25 and 3 miles from a station
- Negative impacts (noise, traffic, crime) reduce prices within 0.25 mile
- Access to rail generally increases property values, but the relationship is non-linear



LITERATURE REVIEW

Debrezion, Pels & Rietveld (2007):

Review of studies on transport accessibility and housing prices

Context :

- Overview of research from different cities and transport systems

Approach :

- Comparative review based on the hedonic price model
- Looked at studies using various research methods and data sources
- Compared how different approaches influence estimated price effects

Findings:

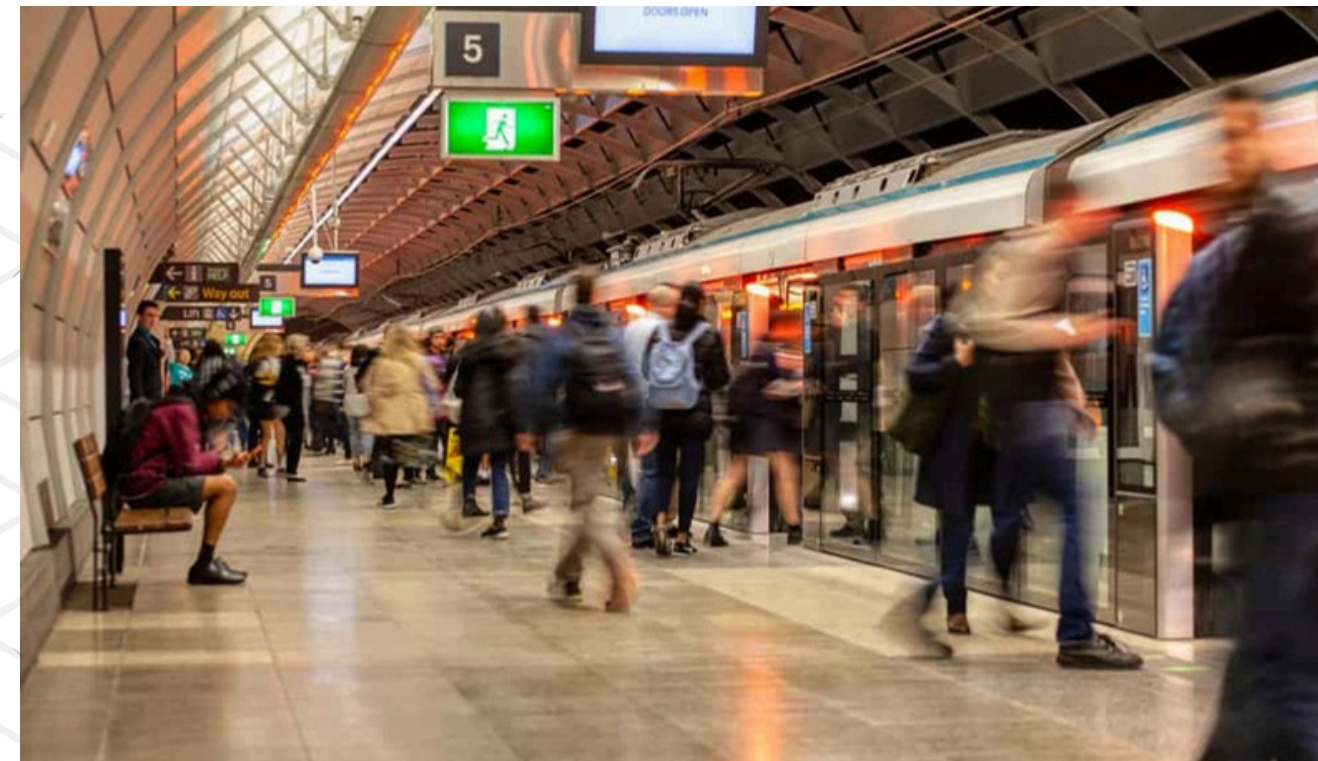
- Living closer to public transport generally increases property prices
- Strongest positive effects for metro and commuter rail systems
- Accessibility is more valuable in city centres with congestion and limited parking
- Studies using advanced methods show smaller but more reliable results

Limitations !

✓ Methodological Differences

X Limited Generalisability

X Location Selection Bias



METHODS : DATA

Dependent Variable

Natural log of House Price

Independent Variable

Distance to Tube Station

Control Variable

- Job Density
- Unemployment Rate
- Hours Worked
- Gross Annual Pay
- Distance to CBD
- Distance to Bus Station
- Time-variant Effects
- Characteristics of Neighbourhoods

MODEL 2 VS LITERATURE

Literature	Our findings	Matchiness
Debrezion, et al. (2007)	proximity to rail is capitalised into housing prices	
Gibbons et al. (2005)	accessibility leads to higher prices in housing	

Benefits:

- 1. Accounts for inherent features of property (lease type, property type, newness, etc.) that Billings, 2011 failed to include

Limitations:

- 1. Does not test the non-linear near-station effect (Bowes, 2001)
- 2. Does not account for neighbourhood socio-economics, proximity to amenities)

CONNECTING TO PREVIOUS RESEARCH

Partly challenges Transport
Driven Valuation
(Gibbons et al, 2005)

Supports Income-based explanations

Extends Inequality debate



Income concentration outweighs
transport effects in shaping
London's housing values

MODEL 7

HDFE Linear regression
Absorbing 2 HDFE groups

Number of obs = 1,126,223
F(13,1125216) = 51594.66
Prob > F = 0.0000
R-squared = 0.6806
Adj R-squared = 0.6803
Within R-sq. = 0.4072
Root MSE = 0.3381

Inprice	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
sq_dist	.0003985	.0005181	0.77	0.442	-.0006169	.0014139
detached_dum	.7101659	.0035354	200.87	0.000	.7032365	.7170952
semi_d_dum	.3368106	.0029631	113.67	0.000	.331003	.3426181
terrace_dum	.1970388	.0028827	68.35	0.000	.1913889	.2026888
freehold	.3387796	.0028681	118.12	0.000	.3331582	.344401
newbuild	.1918595	.0013811	138.91	0.000	.1891526	.1945665
thamesriv_dist	-.0000236	1.18e-06	-19.98	0.000	-.0000259	-.0000213
dist_to_cbd	-.024108	.0011218	-21.49	0.000	-.0263067	-.0219093
bus_distnear	.0002798	3.56e-06	78.52	0.000	.0002729	.0002868
grossannualpay	5.27e-06	3.18e-07	16.55	0.000	4.64e-06	5.89e-06
jobdensity	.5000942	.0079558	62.86	0.000	.4845011	.5156873
hoursworked	.0137021	.0009425	14.54	0.000	.0118549	.0155493
unemployment	-.0184727	.0003432	-53.82	0.000	-.0191455	-.0178
_cons	11.74769	.0404391	290.50	0.000	11.66843	11.82695

Absorbed degrees of freedom:

Absorbed FE	Categories	- Redundant	= Num. Coefs
year	10	0	10
msoa11	985	1	984

Reasonableness:

1. Consistent sensibility (lessen couple meters not worth extra cost of house price)

R-SQUARE VALUE IS 0.68

Factor (square distance) did not improve the model when compared to model 3a and 3b → price distance relationship likely linear

T-VALUE < 2

factor is statistically insignificant

MODEL 7 VS LITERATURE

Literature	Our findings	Matchiness
Bowes et al. (2001)	Close proximity to tube station does not lower house prices	
Baum-Snow et al. (2000) and Billings (2011)	Homes closer to the tube station tends to have higher prices	

Limitations:

- 1.t-value insignificant $\neq$ no curvature
- 2.distance ignores walk time and route quality

MODEL 5B

Inprice	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
walk1	-.0528479	.0055338	-9.55	0.000	-.063694	-.0420018
walk2	-.031469	.0053662	-5.86	0.000	-.0419866	-.0209513
walk3	-.0240432	.0052876	-4.55	0.000	-.0344068	-.0136796
walk4	-.0306887	.0052148	-5.89	0.000	-.0409092	-.0204682
walk5	-.0363908	.0051362	-7.09	0.000	-.0464575	-.0263241
walk6	-.0401605	.0050285	-7.99	0.000	-.0500163	-.0303048
walk7	-.0389128	.0048188	-8.08	0.000	-.0483575	-.0294681
walk8	-.0310504	.0045748	-6.79	0.000	-.0400169	-.0220839
walk9	-.0219594	.0041285	-5.32	0.000	-.0300511	-.0138676
(omitted)						
detached_dum	.6426128	.0035019	183.50	0.000	.6357492	.6494764
semi_d_dum	.3238207	.0029058	111.44	0.000	.3181255	.3295159
terrace_dum	.1974973	.0028164	70.12	0.000	.1919773	.2030173
freehold	.3608621	.0028176	128.07	0.000	.3553396	.3663846
newbuild	.1736442	.0014543	119.40	0.000	.1707939	.1764946
thamesriv_dist	-.0000272	2.62e-06	-10.39	0.000	-.0000323	-.0000221
dist_to_cbd	-.014358	.0026262	-5.47	0.000	-.0195052	-.0092108
bus_distnear	.0002195	4.20e-06	52.26	0.000	.0002112	.0002277
grossannualpay	6.10e-06	3.04e-07	20.10	0.000	5.51e-06	6.70e-06
jobdensity	.5061376	.007602	66.58	0.000	.491238	.5210373
hoursworked	.0125659	.0008959	14.03	0.000	.01081	.0143218
unemployment	-.0191724	.0003292	-58.23	0.000	-.0198177	-.0185271
_cons	11.70941	.0449303	260.61	0.000	11.62135	11.79747

Inprice	Coef.
walk1	-.0528479
walk2	-.031469
walk3	-.0240432
walk4	-.0306887
walk5	-.0363908
walk6	-.0401605
walk7	-.0389128
walk8	-.0310504
walk9	-.0219594
walk10	0

STATISTICAL SIGNIFICANCE

t-value < 5%

NEGATIVE CORRELATION

negative coefficient

DECREASES 2-5% MORE

than reference group walk10

DECREASES 2-5% MORE

than reference group walk10

Need for Non-linearity

- Model 7: mathematical operation - squaring
- Model 5: categorical data (Billings, 2011) & (Bowes, 2001)

MODEL 5B VS LITERATURE

Literature	Our findings	Aligns with
Bowes et al. (2001)	station distances have non-linear relationship with WTP	
Bowes et al. (2001)	Houses nearest to a station show positive correlation to WTP due to negative externalities	
Gibbons et al. (2005) & Debrezion, et al. (2007)	Accessibility correlates to higher property houses	
Baum-Snow et al. (2000) and Billings (2011)	Homes closer to the tube station tends to have higher prices	

SHORTCOMINGS ADDRESSED

Omitted Variable Bias

Baum, 2000:
exclusively used
individual data points

Postcode-unit level,
control for MSOA and
LSOA neighbourhood
levels

Billings, 2011: included
inherent property
feature like property
type

More robust: age to
decide newbuild or
not, type of lease

Reverse Causality

cannot be fully curbed, suggesting advanced
approached incorporating more datasets

Billings, 2011: used a
control-treatment
approach & looked at
houses **sold at least**
twice

Gibbons, 2005:
comparing prices
before and after a
railway development

Correlated Effects

distance to amenities: park, river

paying power of residents: **average**
neighbourhood rate of unemployment, no
education, annual pay, hours worked, etc.

Suggestions

control for air pollution, crime,
noise level, distance to malls

THANK YOU

